A Study on Descriptive Analytics

A
DESCRIPTIVE ANALYTICS ON CLIENT SATISFACTION

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1. Introduction

Client satisfaction refers to the extent to which a client feels that a product, service or overall experience meets expectations. It is important because satisfied clients are more likely to continue using a service, recommend it to others and develop trust in an organization.

Descriptive analytics is the process of organizing and summarizing existing data to understand what has happened. It uses simple statistical measures, tables and charts to reveal patterns. In this project, descriptive analytics is used to examine overall client satisfaction scores on a five-point scale.

The report focuses on the distribution of satisfaction scores rather than predicting future behavior or proving cause-and-effect relationships. The provided histogram reports a **sample size of 500, a mean score of 3.77 and a standard deviation of 0.765.**

Objectives of the Project

- Summaries of the available client satisfaction data.
- Understand the typical level and spread of satisfaction scores.
- Interpret the histogram and its normal-curve overlay.
- Identify practical actions that may help improve client experience.

2. Problem Statement

Organizations need a clear understanding of how satisfied their clients are. When feedback is collected but not summarized properly, it can be difficult to identify the overall satisfaction level and whether low ratings are common or limited to a smaller group.

The problem addressed in this project is: **What does the available distribution of overall client satisfaction scores tell us about the satisfaction level of the respondents, and what actions can be suggested from these findings?**

Scope

- The analysis concerns the overall satisfaction score measured on a scale from 1 to 5.
- The available SPSS output represents 500 observations.
- The report uses descriptive interpretation of the displayed histogram and summary statistics.
- The analysis does not establish why clients gave ratings, because supporting demographic or service-factor data were not provided.

Research Questions

- What is the average overall satisfaction score?
- How widely do scores vary around the average?

- Where are scores concentrated in the histogram?
- What improvement steps are reasonable based on the observed pattern?

3. Data Discussion

The data shown in the supplied SPSS output consists of overall client satisfaction scores on a 1–5 scale. The screenshot identifies the dependent variable as "Overall Customer Satisfaction." The displayed sample size is N = 500.

Available Summary Statistics

Measure	Value shown in SPSS
Number of observations (N)	500
Mean satisfaction score	3.77
Standard deviation	0.765
Measurement scale	1 to 5

The mean of 3.77 is above the midpoint of 3 on the five-point scale, suggesting that satisfaction is generally more positive than neutral. The standard deviation of 0.765 indicates a moderate spread in scores around the mean.

Data limitation: The screenshot does not provide the raw dataset, exact frequency table, sampling method, survey questions or respondent profile.

Therefore, exact percentages for each rating and comparisons between client groups cannot be calculated reliably from the image alone.

4. Methods, Tools and Technologies Used

4.1 Method: Descriptive Analytics

Descriptive analytics summarizes observed data using measures such as the mean and standard deviation and by presenting a histogram. It is suitable for this project because the purpose is to describe the existing satisfaction pattern.

4.2 Statistical Measures

- Mean: the average score across the observations; reported as 3.77.
- Standard deviation: indicates how dispersed scores are around the mean; reported as 0.765.
- Frequency distribution: shows how many observations fall within score intervals.
- Histogram: displays the distribution of scores using adjacent bars.
- Normal-curve overlay provides a visual reference for comparing the histogram with a bell-shaped distribution.

4.3 Tools and Technologies

- IBM SPSS Statistics: used to produce the histogram and summary statistics shown in the screenshot.

- Microsoft Word: used to prepare and format this editable report.
- Histogram and normal curve: used for visual interpretation of the score distribution.

No predictive machine-learning model is claimed in this report because the supplied output is descriptive and does not show model training, validation or prediction accuracy.

5. Workflow

- Define the topic: examine overall client satisfaction.
- Identify the measure: use the satisfaction score on a 1–5 scale.
- Review the available output: note the sample size, mean and standard deviation.
- Summaries of the data: use descriptive statistics to describe its center and spread.
- Visualize the distribution: inspect the histogram and normal-curve overlay.
- Interpret the pattern: identify where scores appear most concentrated and whether lower or higher scores are present.
- Develop recommendations: suggest ways to monitor and improve client experience.
- Document limitations: state what cannot be concluded without raw data and additional variables.

Workflow Summary

Data and SPSS output → descriptive statistics → histogram review → interpretation → recommendations.

6. Procedure

The project was carried out by reviewing the supplied IBM SPSS histogram screenshot and extracting the statistics displayed on it. The procedure can be summarized as follows:

- The overall satisfaction variable was identified from the chart title.
- The measurement scale was read as 1–5.
- The displayed sample size, mean and standard deviation were recorded.
- The bar heights were reviewed visually to understand the general distribution.
- The normal curve was treated as a visual comparison rather than proof that the data is perfectly normal.
- The findings were converted into plain-language observations and practical recommendations.

→ Descriptive Statistics → Frequencies or Descriptives to obtain summary measures, and Graphs → Legacy Dialogs → Histogram to create a histogram. The exact menu may vary by SPSS version.

Quality Considerations

- Check that scores fall within the allowed 1–5 range.
- Confirm that missing values are handled consistently.
- Use the original dataset to verify exact frequencies before reporting percentages.
- Keep conclusions proportional to the evidence available.

7. Results / Descriptive Model

The main result is the SPSS histogram of overall client satisfaction scores. The bars are concentrated mainly in the middle-to-higher range, particularly around scores near 3, and the chart shows fewer observations at the lower end of the scale than around the central and higher score ranges.

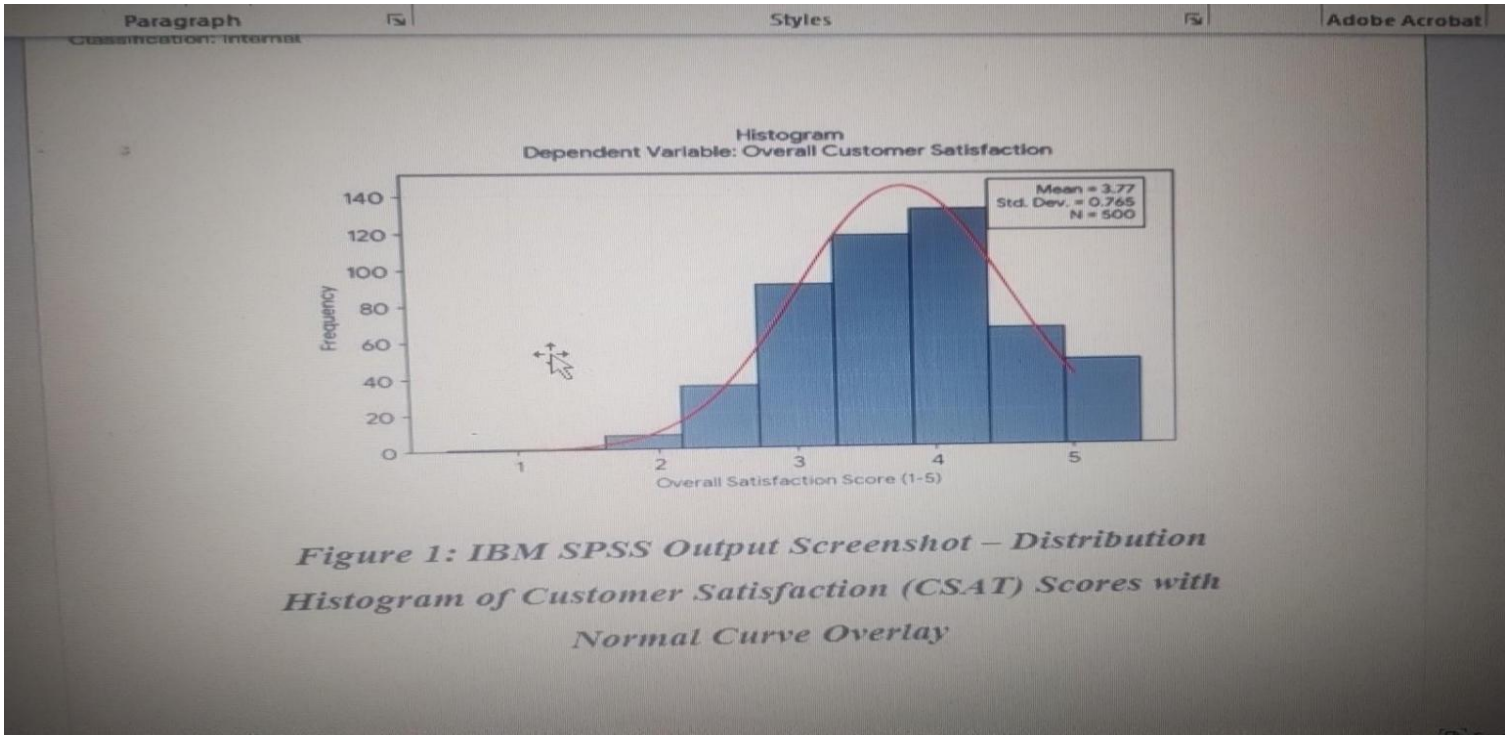


Figure 1. SPSS histogram of overall customer satisfaction scores with normal-curve overlay

Key numerical results: Mean = 3.77; standard deviation = 0.765; N = 500. On a five-point scale, the average is above the midpoint, indicating a generally favourable overall satisfaction level. However, the presence of scores across a range of values suggests that experiences are not identical for all respondents.

The histogram is a descriptive model of the observed scores. It is not a predictive model and does not estimate future satisfaction.

8. Insights and Observations

8.1 Overall Satisfaction Is Generally Positive

The mean score of 3.77 is above the scale midpoint of 3. This suggests that respondents, on average, reported more positive than neutral satisfaction.

8.2 Scores Show Some Variation

The standard deviation of 0.765 indicates that scores are not all clustered at the same value. Some clients report lower satisfaction while others report higher satisfaction.

8.3 Most Scores Appear in the Middle-to-Higher Range

Visually, the tallest bars appear around the scores near 3–4, with additional responses toward the higher end. This indicates that the distribution is centered more on moderate-to-good satisfaction than on very low satisfaction.

8.4 A Normal Curve Is Only a Reference

The curve helps compare the observed bars with a bell-shaped pattern. Visual similarity alone does not confirm normality; formal tests and the original data would be needed for a stronger assessment.

8.5 Important Limitation

The chart alone cannot show which service factors drive satisfaction, whether differences are statistically meaningful, or whether results represent all clients. Those questions require raw data and relevant additional variables.

9. Recommendations / Suggestions

- Investigate lower ratings: review comments and contact dissatisfied clients, where appropriate, to understand specific issues.
- Maintain what works: identify service practices associated with positive feedback and keep them consistent.
- Collect feedback regularly: use a short, clear satisfaction survey after key interactions.
- Track key service factors: include response time, service quality, staff behavior, value for money and issue resolution in future surveys.
- Compare groups carefully: if suitable data is available, compare satisfaction by service type, location or client category while protecting privacy.
- Use a simple dashboard: monitor mean score, rating frequencies and the proportion of low ratings over time.

- Close the feedback loop: communicate improvements made in response to client feedback.

- Validate findings analyses the original dataset and check missing values before making business-wide decisions.

10. Conclusion

This project applied descriptive analytics to the available client satisfaction output. The supplied SPSS histogram summarizes 500 observations on a 1–5 satisfaction scale and reports a mean of 3.77 with a standard deviation of 0.765. The average is above the scale midpoint, and the histogram visually indicates that responses are concentrated mainly in the middle-to-higher range.

The results suggest a generally favorable satisfaction level, while the spread of scores indicates that some clients may have fewer positive experiences. Organizations should therefore maintain strong service practices, investigate lower ratings and collect more detailed feedback to identify improvement areas.

The analysis is limited to the summary information visible in the screenshot. Access to the raw dataset would allow exact rating percentages, additional descriptive measures and comparisons across client groups. The project demonstrates how simple histogram and summary statistics can support evidence-informed decisions without making unsupported predictive claims.

References

- IBM Corp. IBM SPSS Statistics: Descriptive Statistics and Histogram output (screenshot supplied for this project).
- Project dataset summary displayed in the supplied histogram: Overall Customer Satisfaction, N = 500, Mean = 3.77, Std. Dev. = 0.765.

Appendix: Key Results briefly

Item	Summary
Topic	Descriptive Analytics on Client Satisfaction
Sample size	500
Mean	3.77 out of 5
Standard deviation	0.765
Overall interpretation	Generally positive average; variation remains
Analysis type	Descriptive, not predictive